

Muhammad Ahmad Ali

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EDUCATION

Forman Christian College (A Chartered University)

BS (Hons) Computer Science; GPA 3.356

Lahore, Pakistan

Oct 2021 – July 2025

EXPERIENCE

Saailverse

Lahore, Pakistan

Associate Software Engineer

May 2025 – Present

- Designed a Shopify-integrated 3D suit configurator, researching and evaluating multiple integration strategies and taking architectural decisions, identifying limitations in Shopify around variant scalability and dynamic pricing, and designing a Draft Orders-based checkout flow to support fully dynamic configurations that **fully overcomes shopify's 100-variant constraint** and supports **infinite number** of configurable options.
- Developed **15+** production-ready e-commerce mobile screens in React Native, implementing REST API integrations for dynamic data fetching while meeting all UI and functional requirements provided.
- Built a full-featured back-office management application in React for administrative operations that **reduced manual effort by 35%**, implementing end-to-end CRUD operations for seller and buyer accounts, user suspension with corresponding in-app access restriction logic, and delivery tier management system based on client defined requirements and corresponding schemas.
- Supported the development of a 360° virtual tour simulator using Three.js and React, matching desired UI designs, troubleshooting bugs, improving application functionality and supporting ongoing feature development.
- Contributed to a cross-platform healthcare application by developing frontend interfaces using React Native, integrating RESTful APIs for real-time data synchronization.
- Collaborated with a cross-functional team to deliver user-facing dashboards while developing modular, reusable components, maintaining clean code architecture and implementing CRUD operations, data validation systems, and multi-step workflow components across multiple product features as per requirements.

Saailverse

Lahore, Pakistan

Intern

Jan 2025 – Apr 2025

- Refactored more than **20+ React components** from JavaScript to TypeScript and assisted integrating backend APIs, improving code maintainability before MVP delivery and worked on creating documentation for a few components in the project using Gitbooks
- Went through a short training on AI, studying deep learning architectures (ANN, RNN, LSTM, GRU, Transformers) through Jay Alammar's resources and "Attention Is All You Need" paper, then assisted in healthcare chatbot developed by implementing data scraping from **more than 10** different healthcare sites and enabling training on the extracted content to support accurate, query-based responses.

Punjab Safe Cities Authority

Lahore, Pakistan

Final Year Project Collaboration

Oct 2025 – Nov 2025

- Designed and trained ML models for AI-based accident, identification and classification. Worked on enhancing model accuracy by various techniques.
- Explored potential integration with Safe City surveillance infrastructure, assessing feasibility for large-scale deployment.

PROJECTS

Crash AnalytiX (Final Year Project)

- Developed a real-time traffic accident detection system using PyTorch, OpenCV, YOLOv12, DeepSORT, achieving 47 FPS throughput and 0.019s latency.
- Integrated multiple modules including accident detection, severity assessment, license plate recognition, and object identification, attaining an average mAP50 of 92.8 across all modules.
- Implemented a feedback loop that reduced false positives by 23%, enhancing model robustness and reliability over time.
- Deployed the system as a full-stack web application with a React frontend and Flask backend, supporting real-time inference, automated metadata storage and pdf report generation.

Portfolio (React, GSAP, React Three Fiber, Threejs, React Three Drei, TailwindCSS) [↗](#)

- Developed an interactive portfolio website demonstrating my work and technical expertise using React with GSAP for smooth animations and scroll-triggered effects, integrating Three.js for 3D elements and interactive canvas to showcase projects.

PROGRAMMING SKILLS

- Development:** React, GSAP, Tailwind CSS, Threejs, React Three Fiber, React Three Drei, React Native, Express, Node, MongoDB.
- Misc:** SQL, Python, Wordpress, Html, CSS, GitHub, MS Office.